

The diagram illustrates the geometric construction of a hyperbola. A circle is centered at point A . A vertical line segment AT is drawn, with T above A . A point C is located on the circle. A line segment CT is drawn, and a point B is marked on the circle such that AB is perpendicular to CT . A line segment BT is drawn. A point R is located on the line segment AT . A line segment CR is drawn, and a point S is marked on the line segment CT . The diagram includes several angles and segments labeled with numbers 1, 2, 3, 4, indicating specific geometric relationships and constructions.

Dit is reeds bewys dat $\hat{C}_1 = \hat{T} = \hat{A}_1 = \hat{A}_2 = \hat{C}_2 = \hat{S}_1 = x$, en dat $TC = CS$.

(4)

(b) Vervolgens, bepaal $\frac{\text{Oppervlakte } \triangle ABC}{\text{Oppervlakte } \triangle ATR}$.

(4)

[8]